

CLASS
6

QUESTION
PAPER SET

A



**SOF INTERNATIONAL
MATHEMATICS OLYMPIAD
2019-20**

DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Total Questions: 50 | Time: 1 hr.

Guidelines for the Candidate

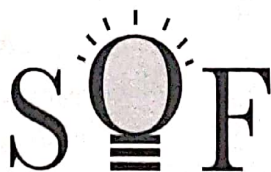
1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. Write your **Name, School Code, Class, Section, Roll No.** and **Mobile Number** clearly on the **OMR Sheet** and do not forget to sign it. We will share your marks / result and other information related to SOF exams on your mobile number.
3. The Question Paper comprises four sections:

Logical Reasoning (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)

Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.

4. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
5. There is only ONE correct answer. Choose only ONE option for an answer.
6. To mark your choice of answers by darkening the circles on the OMR Sheet, use **HB Pencil** or **Blue / Black ball point pen** only. E.g.
Q.16: Rahul bought 4 kg 90 g of apples, 2 kg 60 g of grapes and 5 kg 300 g of mangoes. The total weight of all the fruits he bought is _____.
A. 11.450 kg B. 11.000 kg C. 11.350 kg D. 11.250 kg
As the correct answer is option A, you must darken the circle corresponding to option A on the OMR Sheet.
7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Please fill in your personal details in the space provided on this page before attempting the paper.

16. ☒ A ☐ B ☐ C ☐ D



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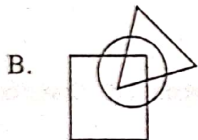
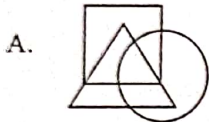
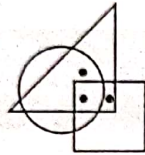
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LOGICAL REASONING

1. If in a certain code language, 'NATION' is written as 'OZUHPM', then how will 'REASON' be written in the same language?

A. SBD RNP
B. SDB RNM
C. QFZ TNO
D. SDB RPM

2. Select a figure from the options which satisfies the same conditions of placement of the dots as in the given figure.

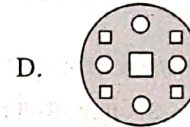
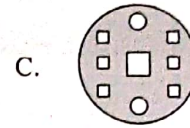
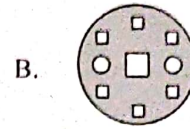
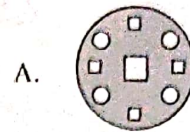
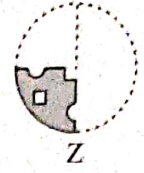
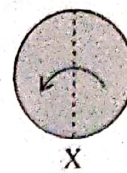


3. Which of the following sets of letters will continue the given letter series?

ARC, BSD, DUF, GXI, ?

A. KBM
B. HBM
C. JAM
D. KAN

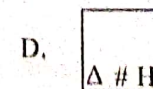
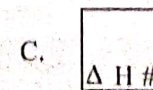
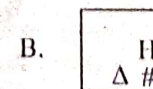
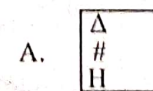
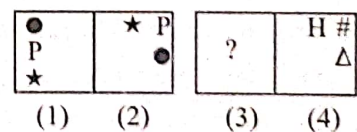
4. The given question consists of a set of three figures X, Y and Z which shows the folding of a piece of paper. Fig. (Z) shows the manner in which the folded paper has been cut. Select a figure from the options which shows the unfolded form of Fig. (Z).



5. Umang walks 20 m towards South. Then he turned left and walks 30 m. Again, he turned left and walks 20 m. Then he turned right and walks 15 m. How far is he now from his initial position?

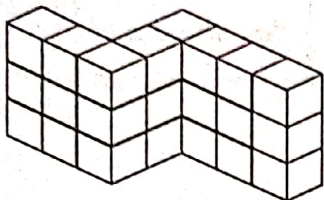
A. 30 m
B. 50 m
C. 35 m
D. 45 m

6. There is a certain relationship between figures (1) and (2). Establish the same relationship between figures (3) and (4) by selecting a suitable figure from the options which will replace the (?) in Fig.(3).



7. The position of how many letters in the word EXPLOSION will remain unchanged, when the letters of the word are arranged alphabetically?
- A. None
B. One
C. Two
D. More than two

8. How many cubes are there in the given figure?



- A. 24
B. 20
C. 21
D. 22
9. If 1 is subtracted from the tens digit and 1 is added to the unit digit in each of the given numbers, then which of the following numbers will become the greatest?

381 376 387 368 328

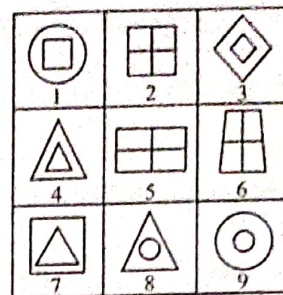
- A. 381
B. 376
C. 387
D. 328
10. Select the correct water image of the given combination of letters and numbers.

S W I M I 9 2 5

- A. 2 M I I W 0 5 2
B. 2 M I W I 0 5 2
C. 2 M I W I 2 0 2
D. 2 I M W I 0 2 5
11. If Karan is the brother of Vijay, Sneha is the daughter of Vijay, Bharti is the sister of Karan and Atul is the brother of Sneha, then how is Karan related to Atul?

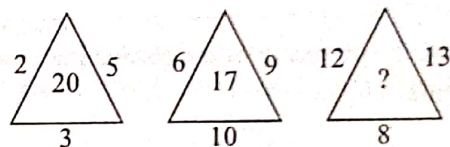
- A. Uncle
B. Brother
C. Father
D. Cousin

12. Group the given figures into three classes on the basis of their identical properties using each figure only once.



- A. 1,7,9; 2,5,6; 3,4,8
B. 1,7,8; 2,5,6; 3,4,9
C. 1,2,4; 3,5,7; 6,8,9
D. 1,2,8; 3,4,6; 5,7,9

13. Find the missing number, if same rule is followed in all the three figures.



- A. 281
B. 495
C. 249
D. 511

14. Select a figure from the options in which the given figure is exactly embedded as one of its parts.



- A.
- B.
- C.
- D.

15. Select the odd one out.

- A. MORS
B. EGJK
C. PQW
D. ACFG

MATHEMATICAL REASONING

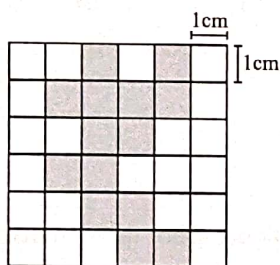
16. If $35.4067 = \frac{3}{P} + 5Q + 4R + \frac{6}{S} + 7T$, then the value of $2P + 4Q + 5R + 7S + 3T$ is

A. 74.003
B. 704.5003
C. 7004.703
D. 7004.7003

17. Find the measure of the smaller angle formed by the hour hand and the minute hand of a clock at 8 O' clock.

A. 160°
B. 240°
C. 90°
D. 120°

18. Find the perimeter of the shaded region in the given figure.



A. 30 cm
B. 32 cm
C. 26 cm
D. 28 cm

19. A hall is 37.5 m long and 11.2 m wide. Its floor is to be covered with rectangular tiles of size 15 cm by 7 cm. Find the total cost of tiling the hall at the rate of ₹ 1.25 per tile.

A. ₹ 25700
B. ₹ 50000
C. ₹ 45000
D. ₹ 48500

20. Arjun is facing South-East. In which direction will he face, if he takes

(a) $1\frac{1}{2}$ revolution anticlockwise?
(b) $\frac{3}{4}$ revolution clockwise?

(a)

A. North-East
B. South-West
C. South-East
D. North-West

(b)

North-East
North-West
North-East
North-East

21. If $\frac{y}{x} = \frac{3}{4}$, then the value of $\left(\frac{1}{4} + \frac{2x-y}{x+2y}\right)$ is

A. 1
B. 4
C. $\frac{3}{4}$
D. 2

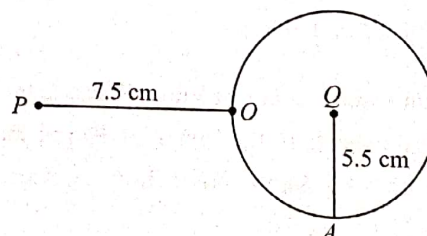
22. If $P = \frac{3}{8} \div \frac{15}{16}$, $Q = 3 \div [(8 \div 15) \div 16]$, $R = [3 \div (8 \div 15)] \div 16$ and $S = 3 \div 8(15 \div 16)$, then which of the following are equal?

A. P and Q
B. P and S
C. P and R
D. All are equal.

23. Find the sum of 43765 and 6954 by rounding off each number to the nearest hundred.

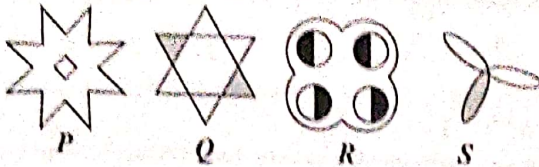
A. 50000
B. 51000
C. 53000
D. 50800

24. O is a point on the circle and P is a point in the exterior of the circle. Length of $\overline{OP} = 7.5$ cm and radius of the circle is 5.5 cm. What will be the length of $\overline{QP}$, if Q is the centre of the circle?



A. 5.5 cm
B. 13 cm
C. 7.5 cm
D. 13.5 cm

25. Which of the following figures has at least one line of symmetry?



- A. Only P
B. Only P and Q
C. Only P, Q and S
D. P, Q, R and S
26. Which of the following ratios is not equal to 3 : 8 in its simplest form?
A. 207 : 552
B. 261 : 696
C. 141 : 392
D. 153 : 408
27. Which of the following operations satisfies the commutative law for whole numbers?
A. Subtraction and division
B. Subtraction and multiplication
C. Division and multiplication
D. Addition and multiplication

Direction (28-29) : The given table shows the number of sweaters knitted by Mrs Kapoor in seven months.

Months	Number of sweaters
May	
June	
July	
August	
September	
October	
November	

28. How many total sweaters did she knit in these seven months?
A. 42
B. 46
C. 54
D. 47
29. How many less sweaters did she knit in August than in November?
A. 5
B. 7
C. 6
D. 4

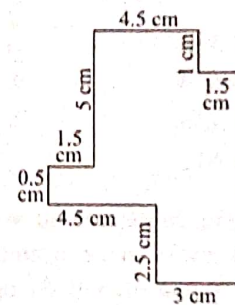
30. The least number which when divided by 12, 16, 24 and 48 leaves a remainder of 3, 7, 15 and 39 respectively, is _____.
A. 16
B. 48
C. 57
D. 39

31. The given table shows the temperature of a city for seven days.

Days	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temperature (in °C)	-6	15	-2	18	12	0	20

- On which day does the temperature increase the most from the previous day?
A. Sunday
B. Thursday
C. Wednesday
D. Tuesday

32. Find the area of the given figure (not drawn to scale).



- A. 25.5 cm²
B. 32.25 cm²
C. 25 cm²
D. 39.75 cm²
33. Select the incorrect option.
A. CLXV - 165
B. MCXLIX - 1149
C. CMLXXIV - 964
D. MCCLVIII - 1258

34. Simplify : $4\frac{1}{3} - 2\frac{4}{9} + 1\frac{3}{5} - \frac{2}{5}$

- A. $3\frac{4}{45}$
B. $1\frac{4}{45}$
C. $3\frac{7}{45}$
D. $1\frac{2}{45}$

35. 1054 is divided into three parts such that 2 times the first part, 3 times the second part and 5 times the third part are equal. Find each of the three parts.

A. 450, 360, 244
B. 350, 456, 248
C. 510, 340, 204
D. 325, 470, 259

EVERYDAY MATHEMATICS

36. Four friends step off together from the same spot. Their steps measure 18 cm, 36 cm, 45 cm and 72 cm respectively. What is the minimum distance each should cover so that all can cover distance in complete steps?

A. 280 cm
B. 340 cm
C. 360 cm
D. 420 cm

37. For a concert, Vishakha spent ₹ 4,74,560 on musical instruments, ₹ 8,12,320 on food, ₹ 3,28,254 on decoration and ₹ 56,480 on advertisement. Find the total amount spent by her.

A. ₹ 21,42,334
B. ₹ 16,71,614
C. ₹ 21,34,136
D. ₹ 15,49,612

38. Niharika walks thrice around a square field of side 22 m. Girish walks twice around a rectangular field of length 10 m and breadth 12 m. Who covers more distance and by how much?

A. Girish, 20 m
B. Niharika, 200 m
C. Girish, 176 m
D. Niharika, 176 m

39. Jennifer bought 6 sarees and 3 shirts. A saree costs twice as a shirt. If each saree costs ₹ 1050.50, then how much total amount did she spend?

A. ₹ 7878.75
B. ₹ 8245.25
C. ₹ 7545.50
D. ₹ 7540.25

40. There are 42 flats in city A and 54 flats in city B. If the cost of each flat in city A and city B is ₹ 2,84,500 and ₹ 2,69,000 respectively, then what is the difference between the cost of all the flats in city A and city B?

A. ₹ 18,45,000
B. ₹ 25,44,000
C. ₹ 27,54,500
D. ₹ 25,77,000

41. 8 years ago, Vinay was $3p$ years old. How old (in years) will he be after 4 years?

A. $3p + 4$
B. $3p + 12$
C. $3p - 4$
D. $3p - 12$

42. An amount of ₹ 5850 is divided into Vishal and Anita in the ratio 4 : 5. Find the share of Anita.

A. ₹ 2600
B. ₹ 3450
C. ₹ 4680
D. ₹ 3250

43. A container is completely filled with four packets of flour weighing $6\frac{3}{8}$ kg, $5\frac{1}{2}$ kg, $2\frac{3}{4}$ kg and $9\frac{1}{2}$ kg.

Find the total weight the container can hold.

A. $26\frac{7}{8}$ kg
B. $24\frac{3}{8}$ kg
C. $24\frac{1}{8}$ kg
D. $24\frac{7}{8}$ kg

44. A rectangular courtyard 1.12 m long and 0.84 m wide is to be paved exactly with square tiles, all of same size. The largest size of the tile which could be used for the purpose is $(n \times 4)$ cm. Find the value of n .

A. 6
B. 8
C. 7
D. None of these

45. Arjun weighs twice as Surbhi and Sejal weighs 5 kg less than Arjun. If the total weight of Arjun, Surbhi and Sejal is 63 kg, then find the total weight of Sejal and Arjun.

A. 54.2 kg
B. 47 kg
C. 49.4 kg
D. 62.4 kg

ACHIEVERS SECTION

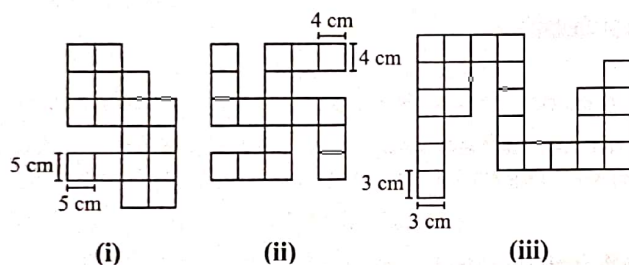
46. Fill in the blanks and select the correct option.

- (i) In 43.295, the digit _____ is at hundredths place.
 (ii) Compare : $11.45 - 3.25 + 12.5$ _____ $15.75 - 4.65 + 9.85$
 (iii) 15.5575 rounded off to the nearest thousandths is _____.

(i) (ii) (iii)

- A. 5 < 15.548
 B. 9 < 15.558
 C. 9 > 15.547
 D. 2 < 15.548

47. Find the perimeter of the following figures (not drawn to scale) and select the correct option.



(i) (ii) (iii)

- A. Perimeter of [(i) + (ii)] = Perimeter of [(ii) + (iii)]
 B. Perimeter of [(i) + (ii)] = Perimeter of (iii)
 C. Perimeter of [(i) + (iii)] > Perimeter of (ii)
 D. None of these

48. Study the given statements carefully and select the correct option.

Statement-I : 7456824 is divisible by 9.

Statement-II : A natural number is divisible by 9, if the sum of the digits of the number is divisible by 3.

- A. Both Statement-I and Statement-II are true.
 B. Both Statement-I and Statement-II are false.
 C. Statement-I is true but Statement-II is false.
 D. Statement-I is false but Statement-II is true.

49. Arrange the following figures in ascending order of their shaded fraction.



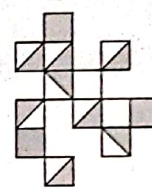
P



Q



R



S

- A. S, P, R, Q
 B. P, S, R, Q
 C. P, S, Q, R
 D. Q, R, S, P

50. Which of the following options is incorrect?

- A. Roman numeral for the largest 3-digit number is CMXCIX.
 B. The smallest 4-digit number formed by using the digits 5, 6, 0, 9 (using each digit only once) is 5069.
 C. Estimated product of 5034 and 295 is 1500000.
 D. Place value and face value are always equal at tens place.

SPACE FOR ROUGH WORK



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